

Section: Particles and Waves

Topic: Refraction of Light

A ray of red light passes from glass into air.

(a)

Explain why the ray bends away from the normal as it enters the air

✔ Light travels **faster in air** than in glass because air is less optically dense.
When light moves from a more dense to a less dense medium, the ray bends **away from the normal**.
This is due to a **decrease in refractive index**.

(b)

The angle of incidence in the glass is 42° and the angle of refraction in air is 65° .
Calculate the refractive index of the glass

Use Snell's Law:

$$n = \frac{\sin i}{\sin r} = \frac{\sin 42^\circ}{\sin 65^\circ}$$
$$= \frac{0.6691}{0.9063} = 0.7385$$

⚠ But this is the refractive index of air *relative to glass*.
To get the **absolute** refractive index of **glass**:

$$n = \frac{1}{0.7385} = 1.35$$

Alternatively, use the version:

$$n_1 \sin i = n_2 \sin r \Rightarrow \frac{\sin 42^\circ}{\sin 65^\circ} = \frac{n_2}{n_1} \Rightarrow n_{\text{glass}} = 1.35$$

(c)

Explain why the value of refractive index would be different for blue light

✔ Blue light has a **shorter wavelength** than red light, and refractive index depends slightly on wavelength (this is called **dispersion**).
Glass refracts **shorter wavelengths more strongly**, so blue light would have a **higher refractive index** than red light in the same medium.

🧠 Revision Tips

- Refraction depends on change in speed of light:
 - From slow to fast = bends **away from normal**
 - From fast to slow = bends **towards normal**

- Snell's Law:

$$n = \frac{\sin i}{\sin r}$$

- Refractive index varies with **wavelength** — this explains the formation of rainbows!